

GE Healthcare

Signa HDx Series 8–Channel to 16–Channel Upgrade

Technical Publication



5166010
Rev 4

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・このサービスマニュアルには英語版しかありません。
- ・GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
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注意：

- 本维修手册仅存有英文本。
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- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

Revision History

Date	Rev	Change Description
14–May–2006	1	Initial Release
6–Jun–2007	2	Complete Revision of manual (ECO 2041415). Section 1: Added information and flowchart for upgrading system with or without B–Port Connector. Added catalog M3335A. Section 2: Added procedures for upgrading site that did not have second cable track or B–Port connector as part of forward production installation. Added cable maps for each upgrade scenario.
22–Sep–2008	3	Updated Section 1–1 to include HDxt (Release 15.x) information.
10–Jun–2009	4	Section 1: Added Catalog offerings M3335BR & M3335BS. Updated upgrade flowchart. Section 2: Updated cable information table and cable maps. Added Section 2–5 for upgrading sites with depopulated 16–Channel Switches.

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SECTION 1 – OVERVIEW

1–1 INTRODUCTION

This manual provides instructions to upgrade a Signa HDx (Release 14.x) or HDxt (Release 15.x) system from an 8–Channel system to 16–Channels.

Note

8–Channel systems are produced in two configurations. This document contains instructions for installing the upgrade for either configuration. Please observe whether your 8–Channel system contains a B–Port connector to determine which instructions to follow.

This Direction covers the following catalogs:

M3335R – 8 Channel to 16 Channel Upgrade Kit
M3335A – 8 Channel to 16 Channel Upgrade 8W8 Cables
M3335BR – 3.0T 16 Channel Switch Upgrade
M3335BS – 1.5T 16 Channel Switch Upgrade

1–1–1 Upgrade Descriptions

For systems **with a 8–Channel B–Port Connector** this upgrade:

- Adds one cable between the 16 Channel Switch and the Image Reject Filter (M3335R)
- Replaces the B8 Bezel with the B16 Bezel (M3335R)

See **Section 2–3** for the upgrade procedure.

For systems **without a B–Port Connector** this upgrade:

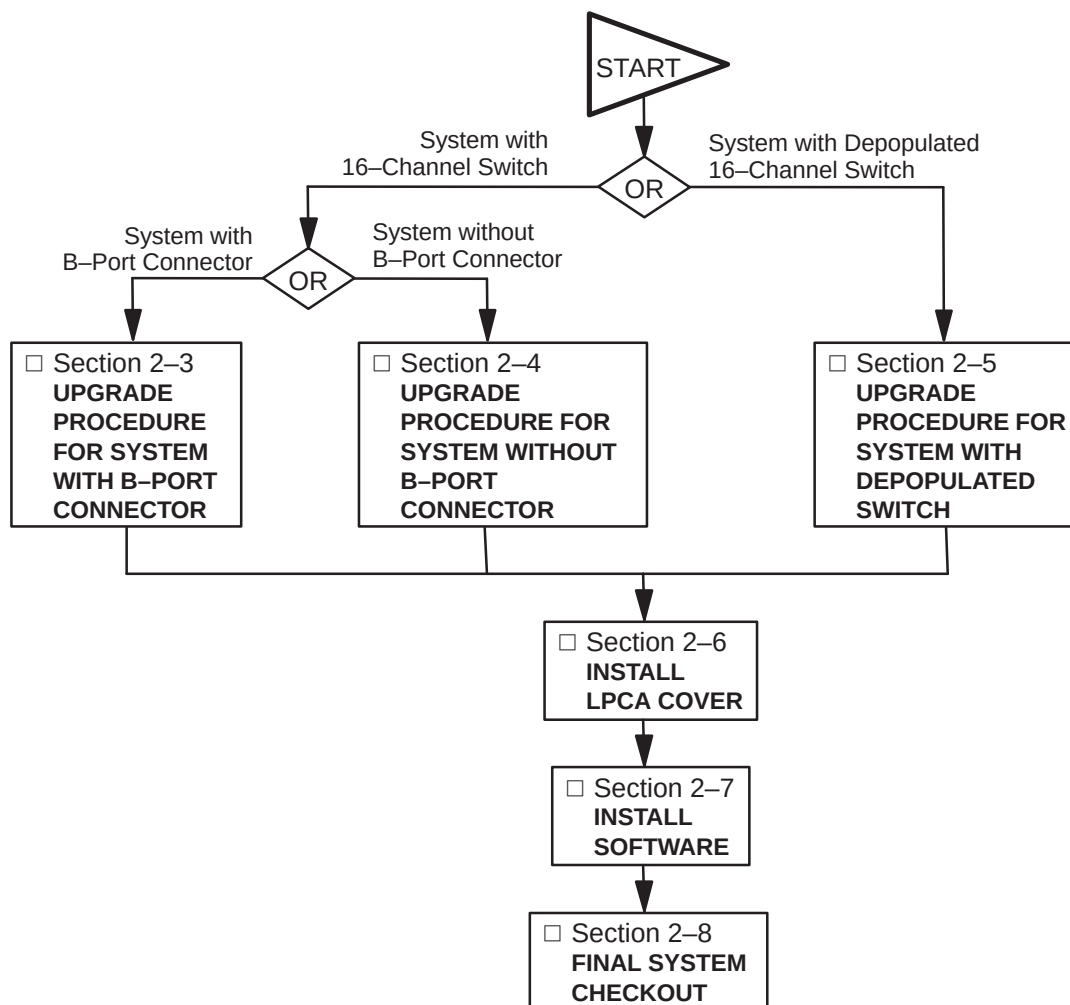
- Adds the B16 Bezel and associated cable track (M3335R)
- Adds two cables between the 16 Channel Switch and the Image Reject Filter (M3335R)
- Adds one cable between the 16 Channel Switch and the Penetration Panel (M3335A)
- Adds one cable between the Penetration Panel and the System Cabinet (M3335A)

See **Section 2–4** for the upgrade procedure.

For systems **with a Depopulated 16–Channel Switch, without an IRF Module, and without a B–Port Connector** this upgrade:

- Adds a 16–Channel Switch, Image Reject Filter (IRF) Module and associated cables (M3335BR or M3335BS)
- Adds the B16 Bezel and associated cable track (M3335R)
- Adds cables between the Penetration Panel, System Cabinet, and 16–Channel Switch (M3335A)

See **Section 2–5** for the upgrade procedure.

1-1-2 Upgrade Flowchart:**1-2 DISCARDED MATERIAL RETURN POLICY**

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1-3 SOFTWARE LOAD FOR UPGRADE

Install the 16-Channel Option key.

1–4 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans–shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at:
<http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following:
Product Locator Support Central Page – http://supportcentral.ge.com/products/sup_products.asp?prod_id=15563

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans–shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1–5 FINAL COORDINATION

1. Replace all covers that have not been previously replaced.
2. Record and enter applicable data into applicable site configuration files and records.
3. Process product locator installation cards for all serialized components added to the system to include product locator card supplied with the upgrade serial number rating plate. Return to:

Product Locator File
P.O. Box 414, W–523
Milwaukee, WI 53201–0414

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

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SECTION 2 – UPGRADE PROCEDURES

This section contains upgrade procedures for upgrading a system with an existing 16–Channel Switch or 16–Channel Depopulated Switch. Refer to upgrade flowchart in Section for guidance.

- LPCA with B–Port Connector – Section 2–3
- LPCA without B–Port Connector – Section 2–4
- Rear Pedestal with Depopulated 16–Channel Switch & without IRF Module, LPCA without second cable track and B–Port Connector – Section 2–5

Select the applicable upgrade procedure for the site being upgraded.

Not all parts provided in M3335R will be installed when performing the upgrade procedures in Section 2–3 since the cable–track and cables were already installed at the initial installation of the system. Discard unused parts per policy stated in Section 1–2.

2–1 CABLE RUN INFORMATON**2–1–1 M3335R and M3335BS**

The table below provides information for connecting the cables provided in M3335R, M3335BR, and M3335BS.

Run Number	Part Number	Connection A	Connection B	Cables to be Installed if System Upgraded has:		
				16–Channel Switch M3335R		Depopulated 16–Channel Switch M3335BS M3335BR
				With B–Port	No B–Port	
1334	5141244	Image Reject #1 J5	16 Ch Sw #1 J22		Yes	Yes
1335	5141362	Image Reject Filter #1 J6	16 Ch Sw #1 J17			Yes
1336	5141244–2	Image Reject #1 J7	16 Ch Sw #1 J24	Yes	Yes	Yes
1337	5141363	Image Reject Filter #1 J8	16 Ch Sw #1 J19			Yes
1338	5141982	Image Reject Filter #1 J4	LPCA–J3			Yes
1339	5141980	Image Reject #1 J1	LPCA–J23		Yes	Yes
1340	5141979	Image Reject #1 J2	LPCA–J22		Yes	Yes
1341	5141980–2	Image Reject #1 J3	LPCA–J21		Yes	Yes

2–1–2 M3335A

The table below provides information on the cables provided in M3335A. A short and long length is provided for each cable Run. Only one of each Run will be installed.

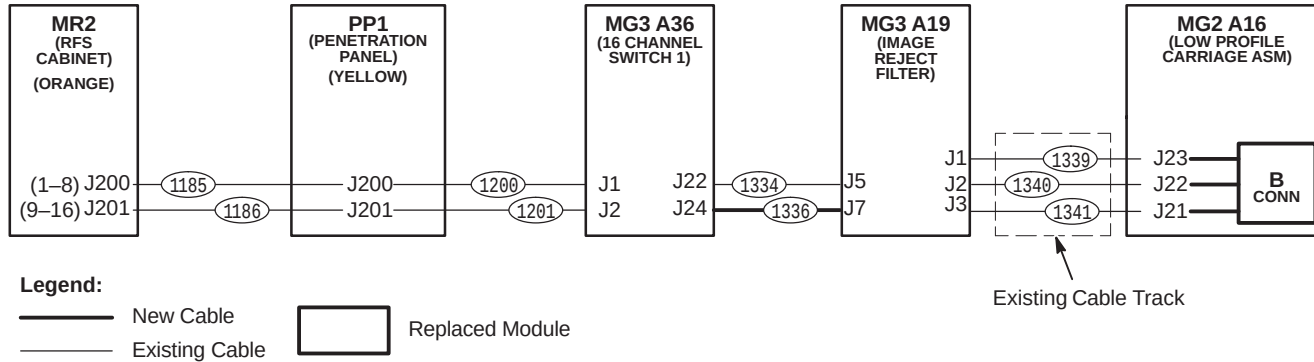
- Run 1186 must be the same length as Run 1185, which is already installed in the Equipment Room.
- Run 1201 must be the same length as Run 1200, which is already installed in the Magnet Room.
- In some early installations of 8–Channel systems, Runs 1217 and 1329 may have been installed. If this is the case for the system being upgraded, the new provided Runs 1217 and 1329 are not installed.
- If Runs 1217 and 1329 are to be installed, they must be the same length as Runs 1186 and 1201.

Run Number	Part Number	Connection A	Connection B
1186	2394869–2 (Short)	Penetration Panel J201	RFS Cabinet J201
	2394869–4 (Long)	Penetration Panel J201	RFS Cabinet J201
1201	5107475–2 (Short)	Penetration Panel J201	16 Ch Sw #1 J2
	5107475–4 (Long)	Penetration Panel J201	16 Ch Sw #1 J2
1217	5113186–9 (Short)	Penetration Panel J78	16 Ch Sw #1 J10
	5113186–10 (Long)	Penetration Panel J78	16 Ch Sw #1 J10
1329	5160765 (Short)	Penetration Panel J78	RFS Cabinet J10
	5160765–2 (Long)	Penetration Panel J78	RFS Cabinet J10

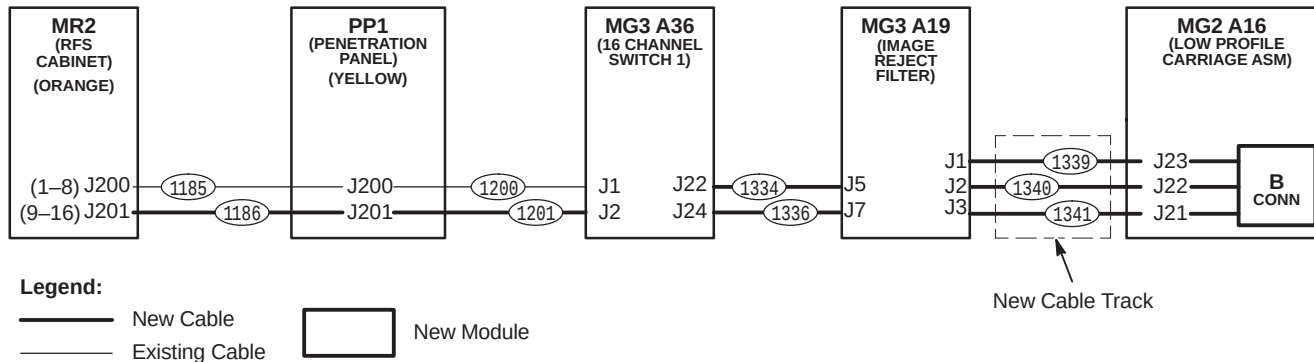
2-2 8-CHANNEL TO 16-CHANNEL UPGRADE INTERCONNECT MAPS

Refer to the applicable cable map below for upgrading site.

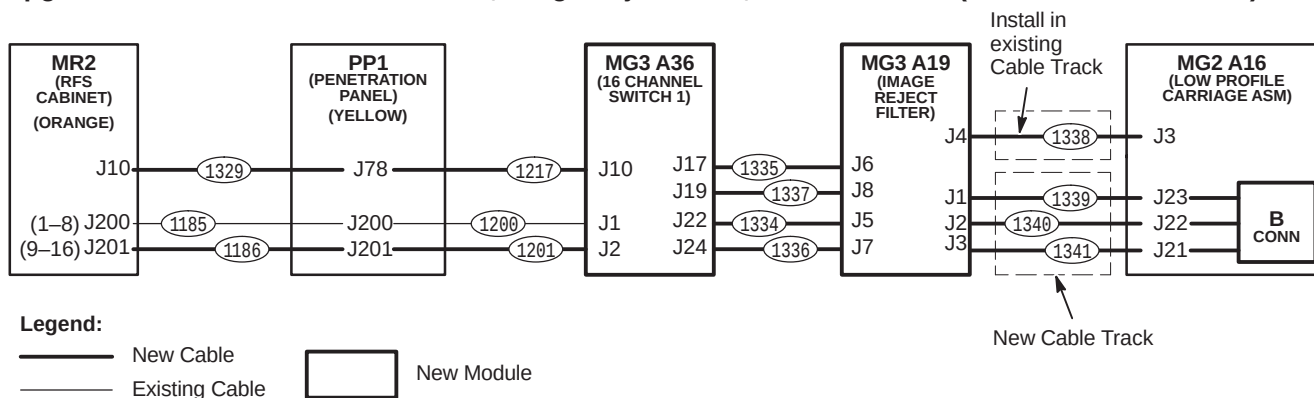
Upgrade with Existing 16-Channel Switch and 8-Channel B-Port (M3335A and M3335R)



Upgrade with Existing 16-Channel Switch, without B-Port (M3335A and M3335R)



Upgrade with New 16-Channel Switch, Image Reject Filter, without B-Port (M3335A and M3335BS)



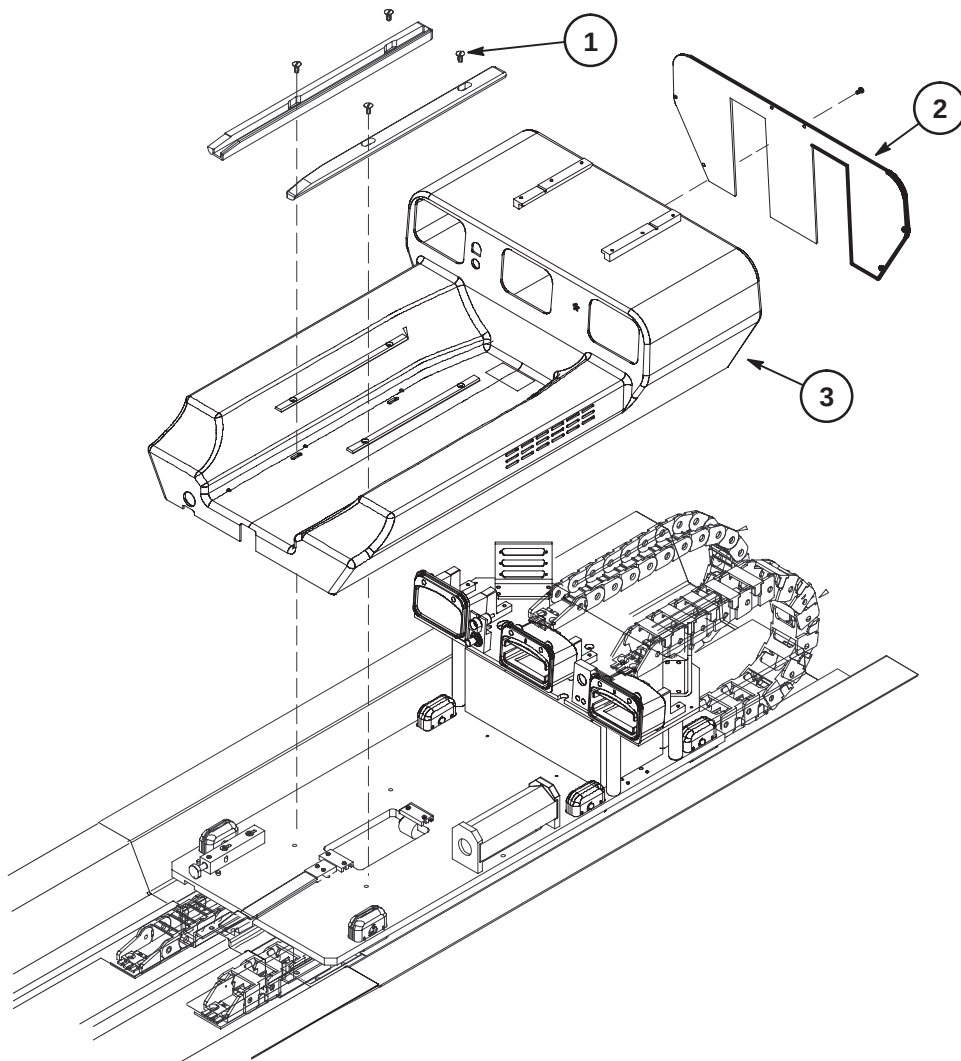
2-3 UPGRADE PROCEDURE FOR LPCA WITH B-PORT CONNECTOR

The upgrade procedures in Section 2-3 include parts from catalogs M3335R and M3335A and will reuse the existing 16-Channel Switch and Image Reject Filter.

The parts consist of the B16-Bezel & cable harness and cable installed in the Rear Pedestal.

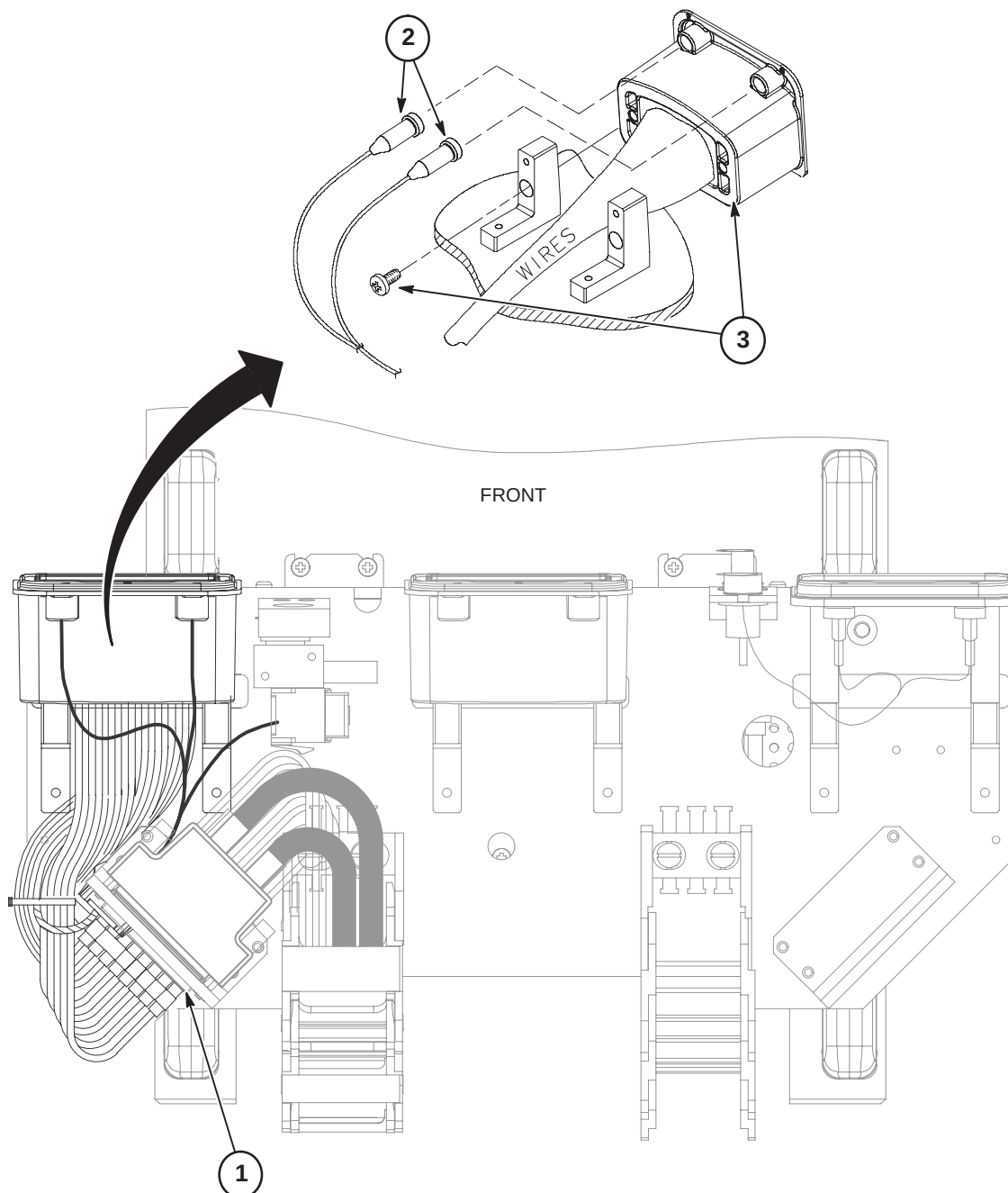
2-3-1 Remove LPCA Cover

1. Unfasten the four screws that attach the cover to the base plate.
2. Unfasten 6 screws and remove Rear Cover (2386346) from LPCA
3. Carefully remove cover from base plate assembly.



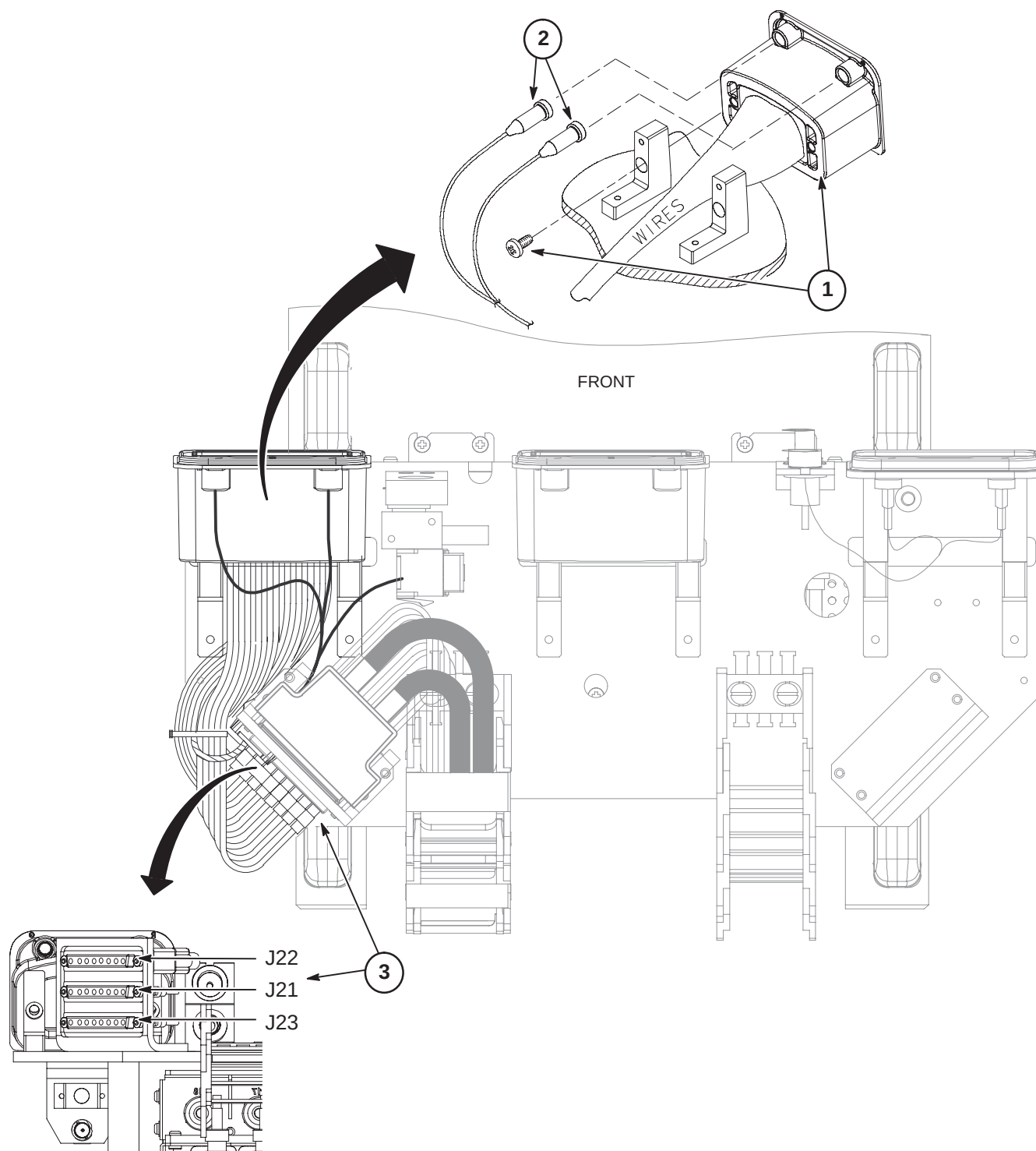
2-3-2 De-Install B8-Bezel From LPCA

1. Disconnect harness from J22 and J23. Save fastening hardware for connecting new harness.
2. Disconnect the 2 LED's inserted into Bezel.
3. Unfasten the two M5 x 20mm Brass Screws and de-install the B8 Connector.



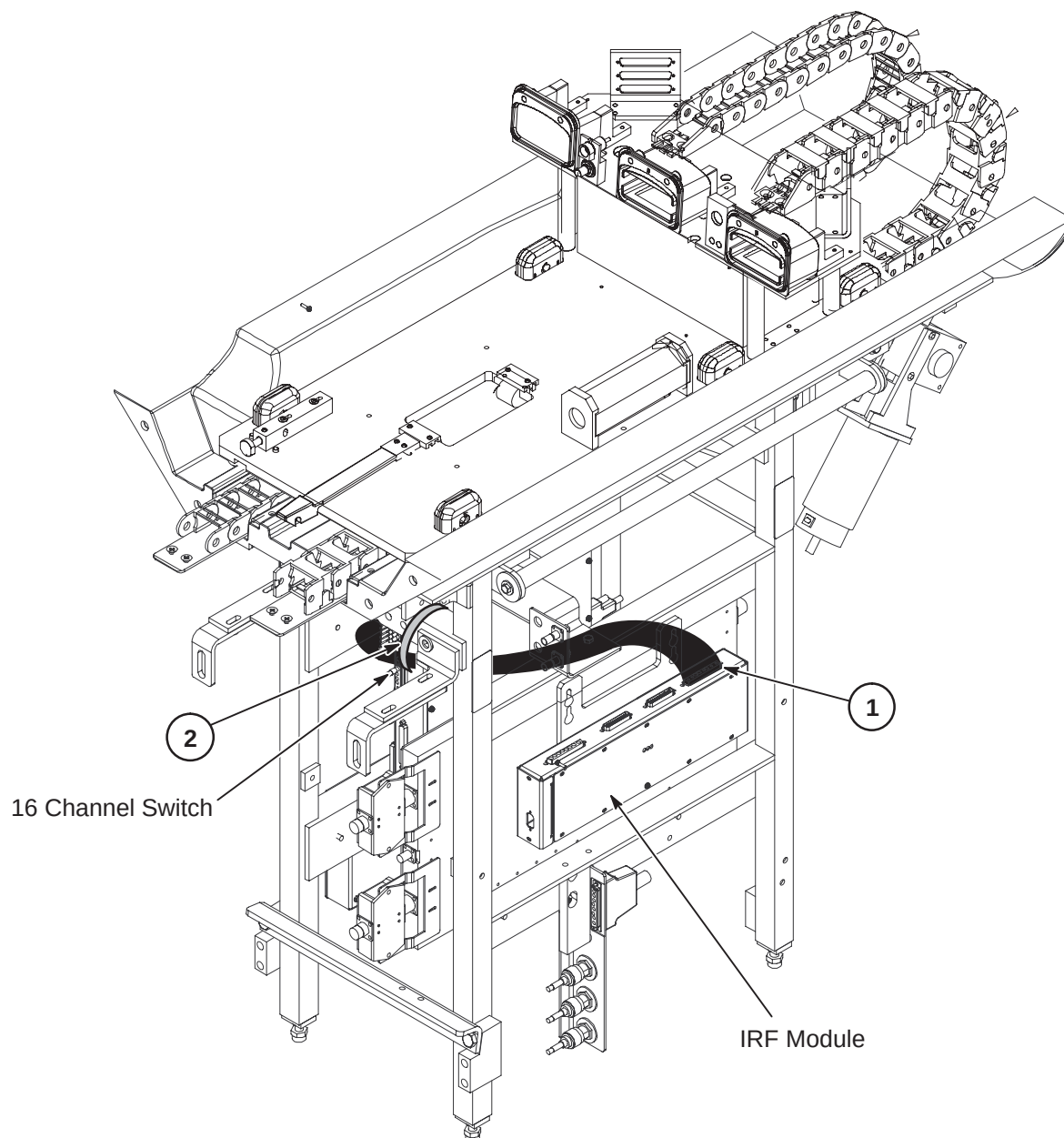
2-3-3 Install B16-Bezel in LPCA

1. Install B16 Bezel and Cable harness (5111020-52) using the two M5 x 20mm Brass Screws previously removed.
2. Insert the 2 LED's into Bezel.
3. Connect harness to J21, J22, and J23 on bracket as marked using previously removed fastening hardware.



2-3-4 Install Run 1336

1. Connect one end of Run 1336 as marked to J7 on IRF module. Route along upper front bracket of Rear Pedestal and connect to 16 Channel Switch J24.
2. Dress and tie-wrap Run 1336 to upper front bracket of Rear Pedestal.



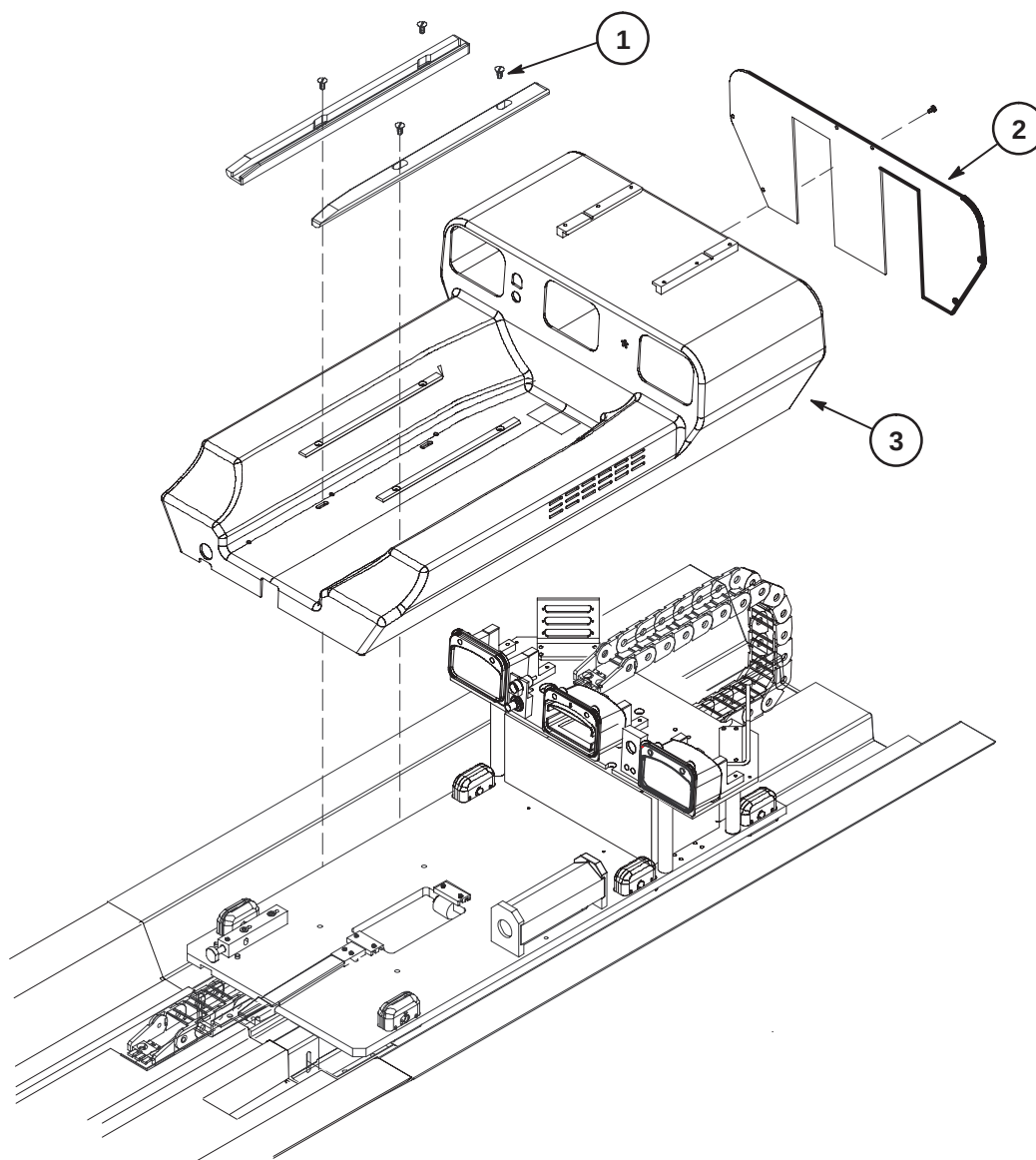
2-4 UPGRADE PROCEDURE FOR LPCA WITHOUT B-PORT CONNECTOR

The upgrade procedures in Section 2-4 include parts from catalogs M3335R and M3335A and will reuse the existing 16-Channel Switch and Image Reject Filter.

The parts consist of the B16-Bezel and cable harness, Cable Track with cables, cables installed in the Rear Pedestal, and cables routed to RFS Cabinet.

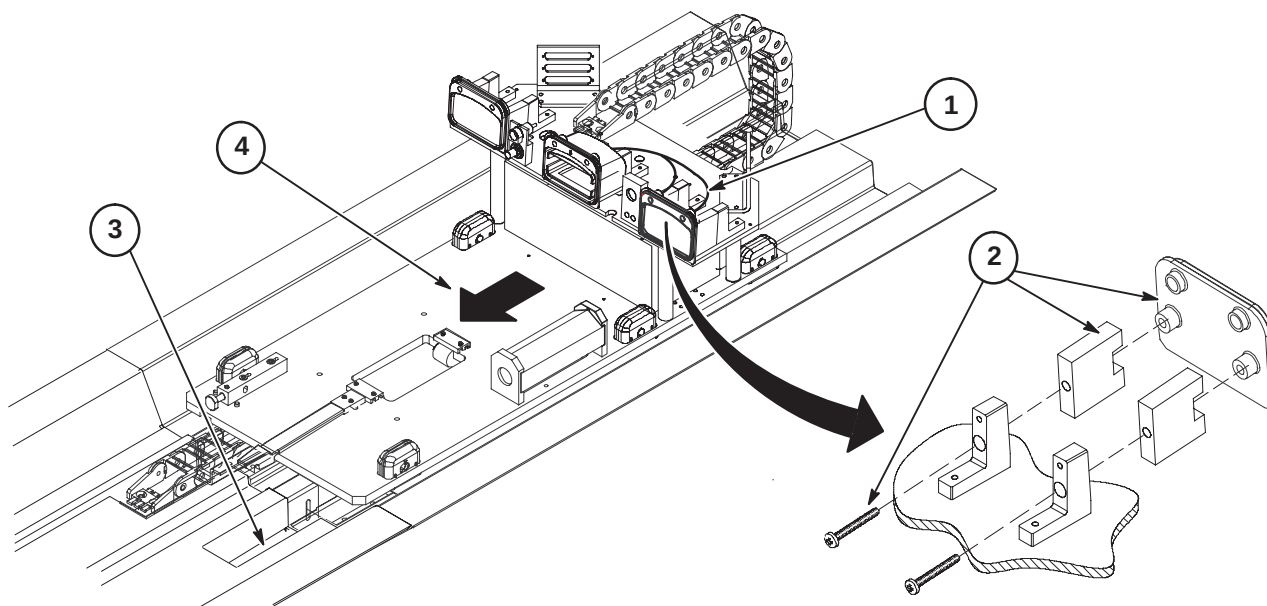
2-4-1 Remove LPCA Cover

1. Unfasten the four screws that attach the cover to the base plate.
2. Unfasten 6 screws and remove Rear Cover (2386346) from LPCA
3. Carefully remove cover from base plate assembly.



2-4-2 De-Install the B-Port Blanking Plate and the Cable Track Cover Plate

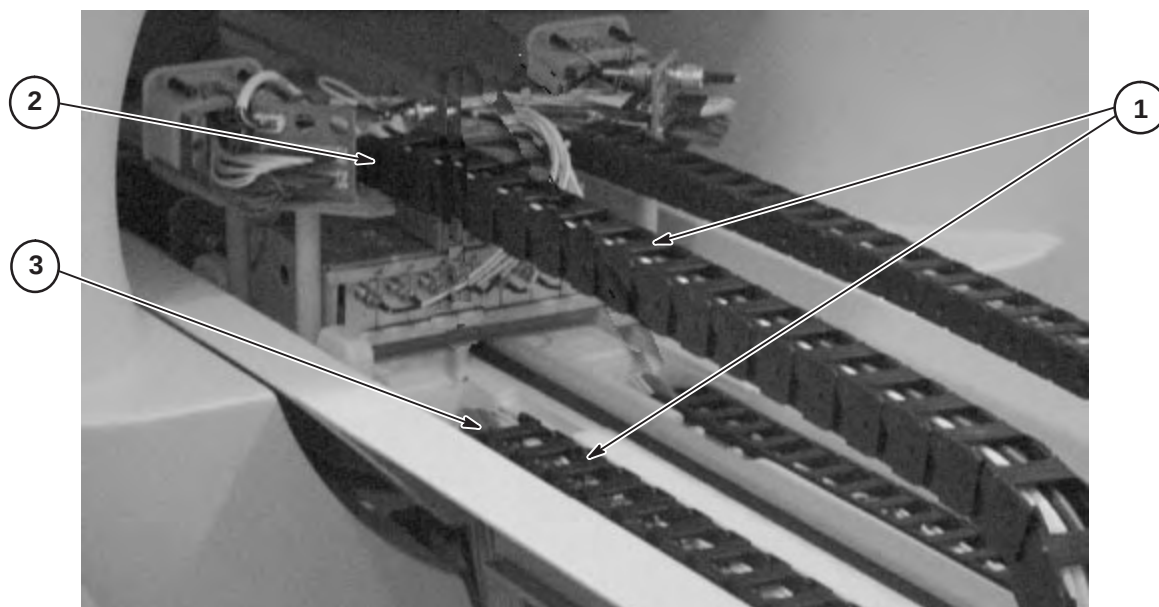
1. Disconnect and retain LED cables from back of Bezel Cover.
2. Unfasten and discard the existing Bezel Cover, Spacer, and M5 x 20mm Screws.
3. Remove and discard Cable Track Opening Cover. Retain screws for installation of new Cable Track.
4. Move LPCA forward into bore so rear of LPCA plate is located just past the cable track opening in the rear of the Front Split Bridge.



2-4-3 Install Cable Track and Runs 1339, 1340, and 1341

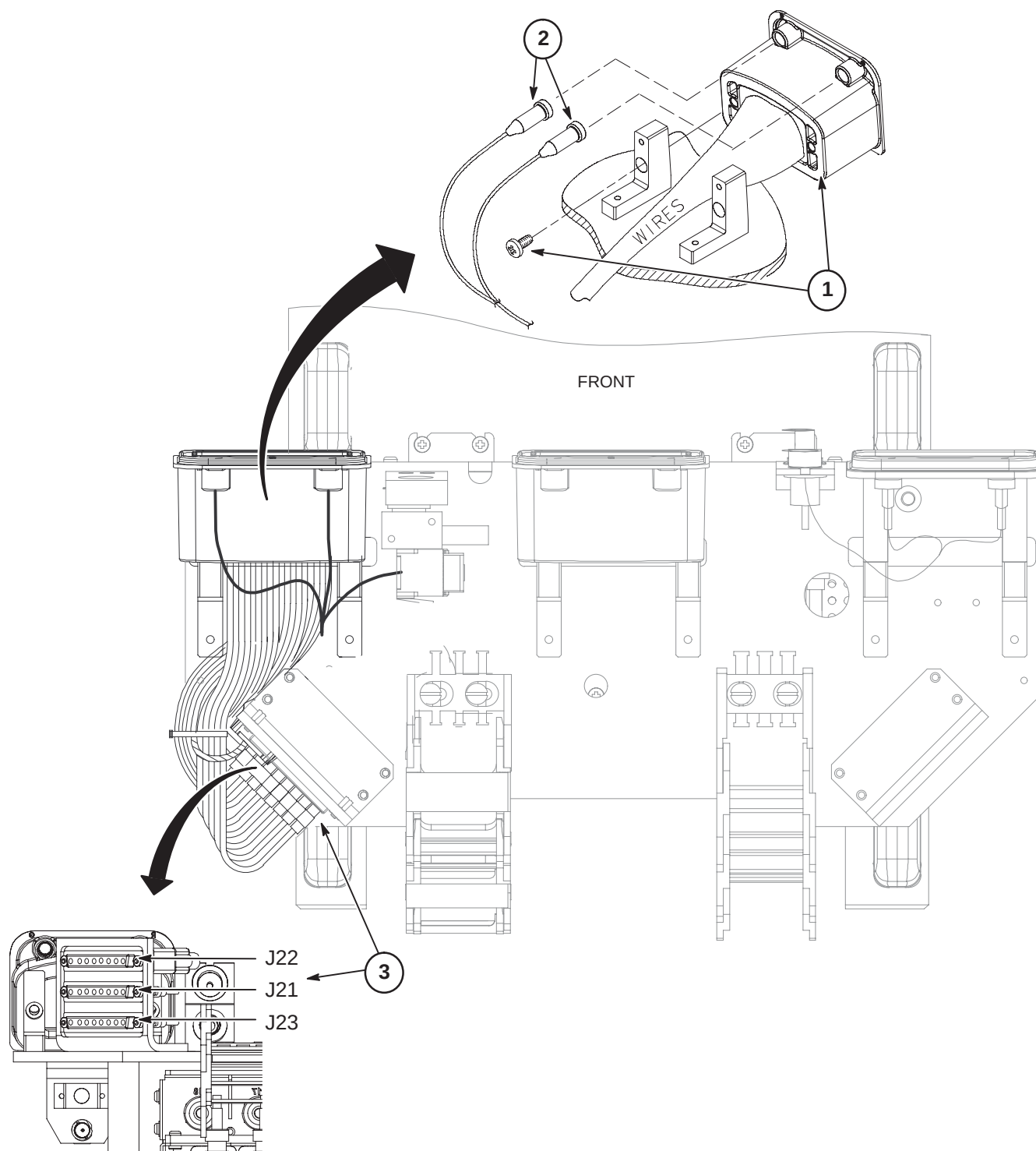
1. Insert the Cable Track assembly into the rear bridge cable track.
2. Remove two screws at location where end of Cable Track attaches to LPCA and apply two drops of LockTite to each screw. Fasten the LPCA end of the cable track to the LPCA with the two screws.
3. Apply two drops of LockTite to the two screws previously removed at the cable track opening in Bridge and attach the other end of the Cable Track.

NOTE: Failure to apply two drops of proper thread locking compound to each cable chain screw before installation could result in screws loosening later and causing spike noise.



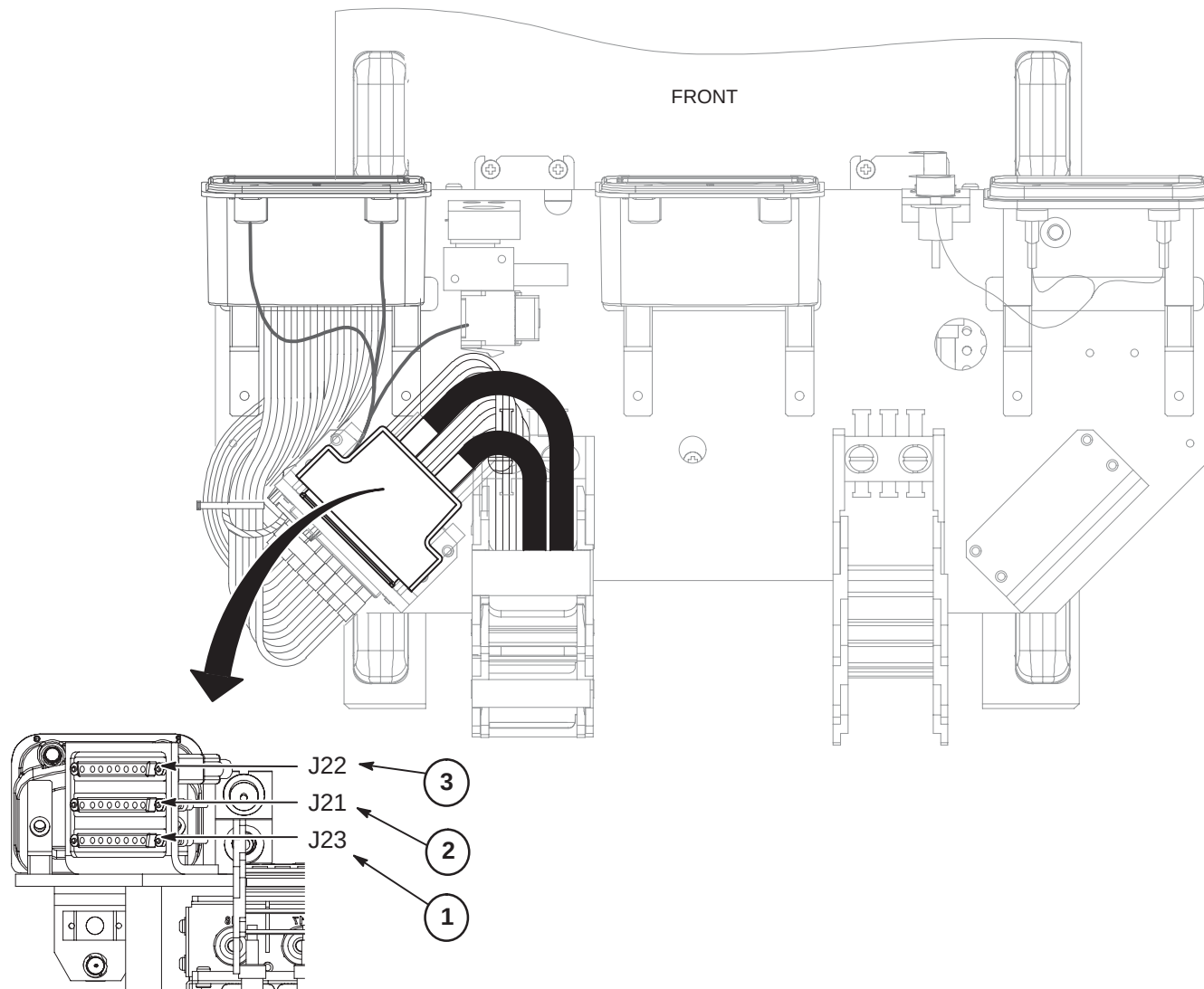
2-4-4 Install B16-Bezel in LPCA

1. Install B16 Bezel and Cable harness (5111020-52) using two M5 x 20mm Brass Screws.
2. Insert the 2 LED's into Bezel.
3. Connect harness to J21, J22, and J23 to bracket as marked using provided nut and standoff on bracket.



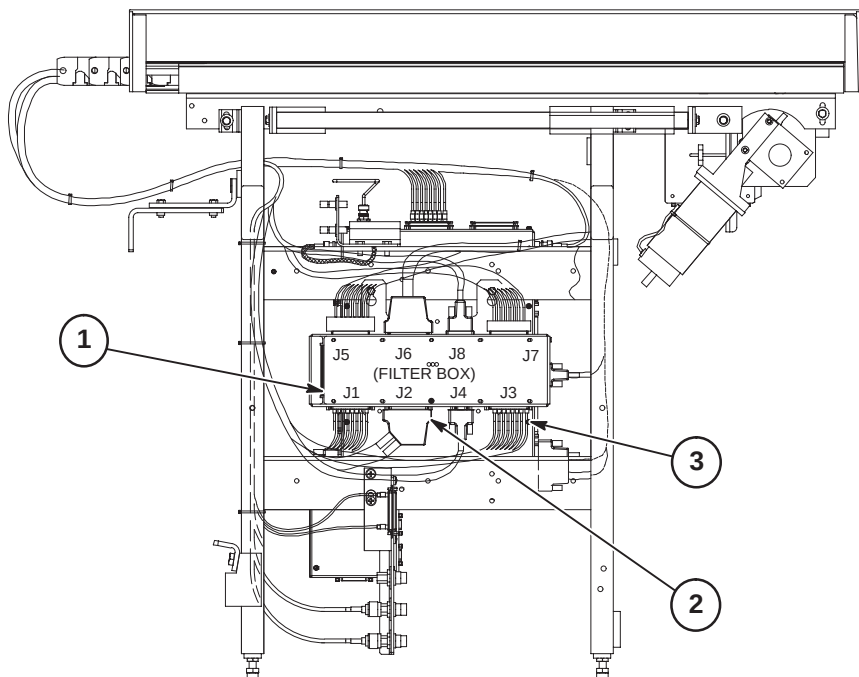
2-4-5 Connect Runs 1339, 1340, and 1341 to B16 Bezel Connectors

1. Connect end of cable run 1339 to J23 on the LPCA bracket.
2. Connect end of cable run 1341 to J21 on the LPCA bracket.
3. Connect end of cable run 1340 to J22 on the LPCA bracket.



2-4-6 Connect Runs 1339, 1340, and 1341 to IRF Module

1. Connect cable run 1339 to J1 on the Image Reject Filter.
2. Connect cable run 1340 to J2 on the Image Reject Filter.
3. Connect cable run 1341 to J3 on the Image Reject Filter.

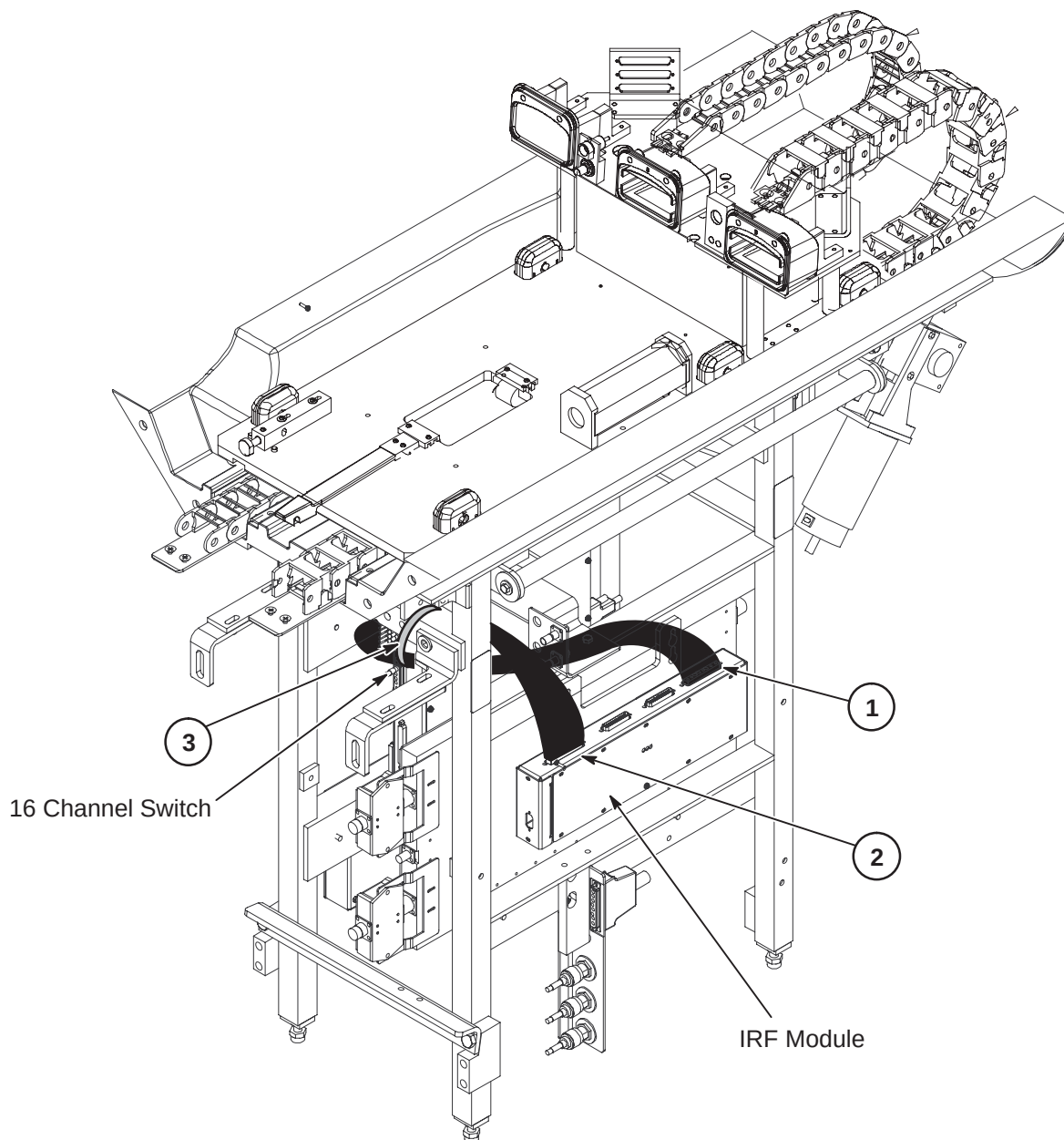
**2-4-7 Install Runs 1186 and 1201**

Refer to applicable cable map in Section 2-2.

1. Connect one end of Run 1186 as marked to J201 on Penetration Panel. Route and connect other end to RFS Cabinet J201.
2. Connect one end of Run 1201 as marked to J201 on Penetration Panel. route and connect other end to 16-Channel Switch J2.

2-4-8 Install Runs 1334 and 1336

1. Connect one end of Run 1336 as marked to J7 on IRF module. Route along upper front bracket of Rear Pedestal and connect to 16 Channel Switch J24.
2. Connect one end of Run 1334 as marked to J5 on IRF module. Route along upper front bracket of Rear Pedestal and connect to 16 Channel Switch J22.
3. Dress and tie-wrap both Runs to upper front bracket of Rear Pedestal.



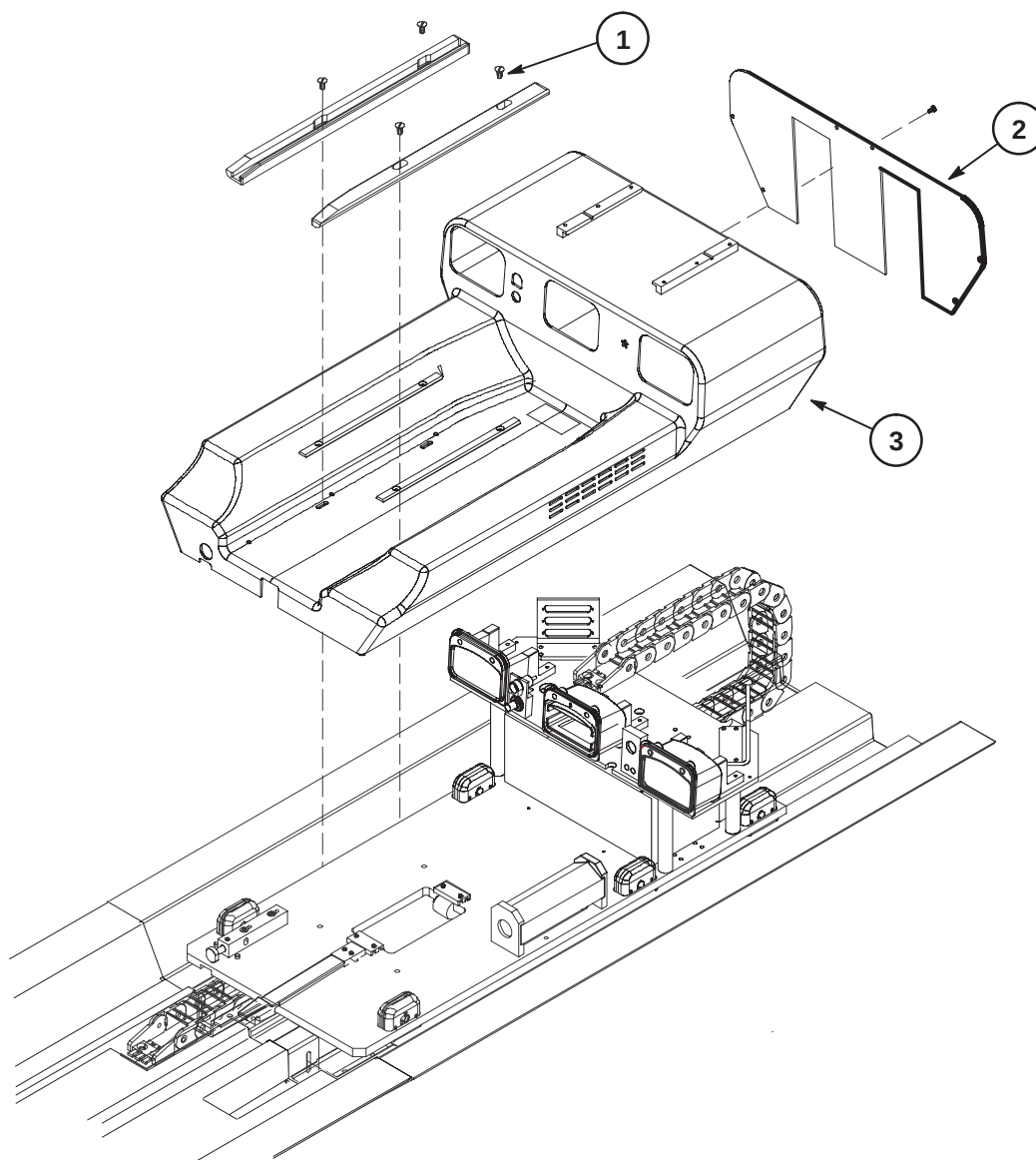
2-5 UPGRADE PROCEDURE FOR REAR PEDESTAL WITH DEPOPULATED 16-CHANNEL SWITCH

The upgrade procedures in Section 2-5 include parts from catalogs M3335BS (1.5T) or M3335BR (3.0T) and M3335A.

The parts consist of the 16-Channel Switch, Image Reject Filter, B16-Bezel and cable harness, Cable Track with cables, cables installed in the Rear Pedestal and existing cable track, and cables routed to RFS Cabinet.

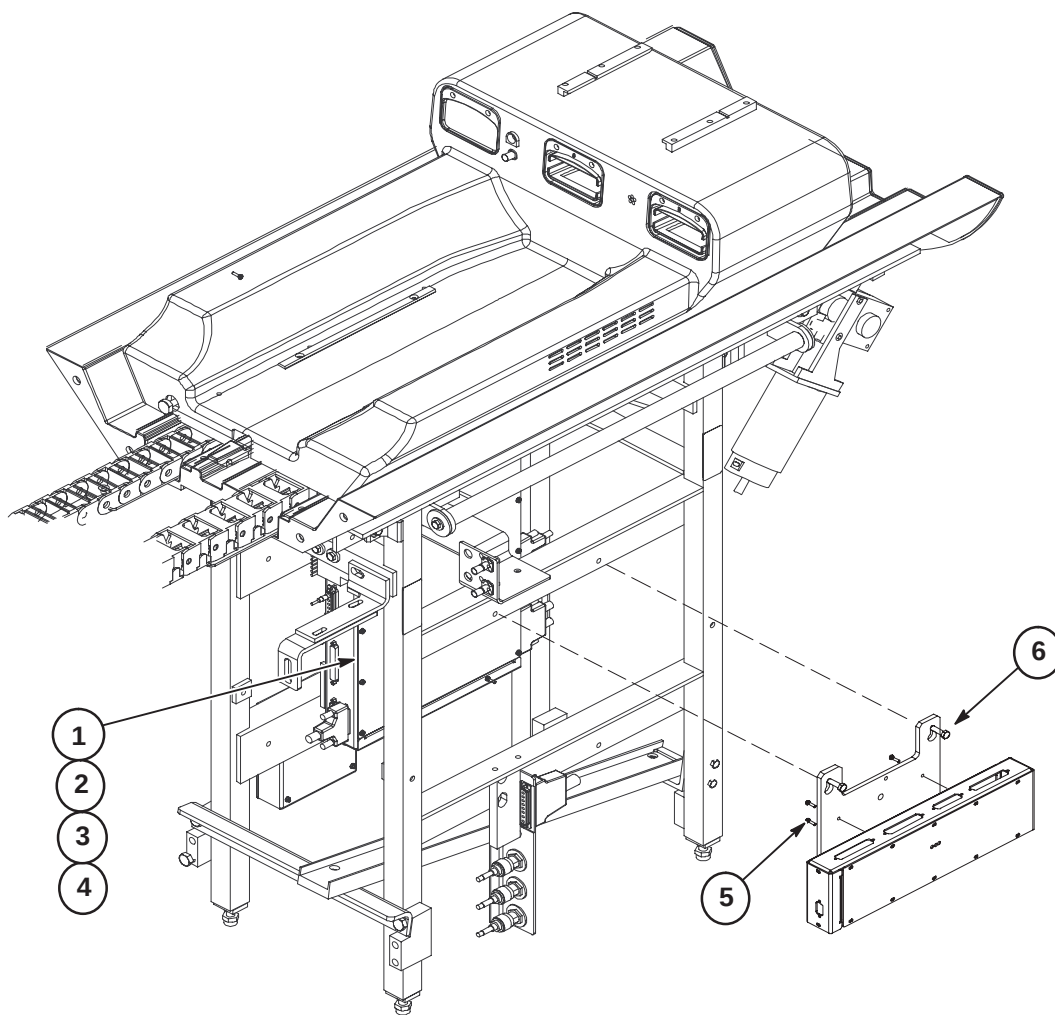
2-5-1 Remove LPCA Cover

1. Unfasten the four screws that attach the cover to the base plate.
2. Unfasten 6 screws and remove Rear Cover (2386346) from LPCA
3. Carefully remove cover from base plate assembly.



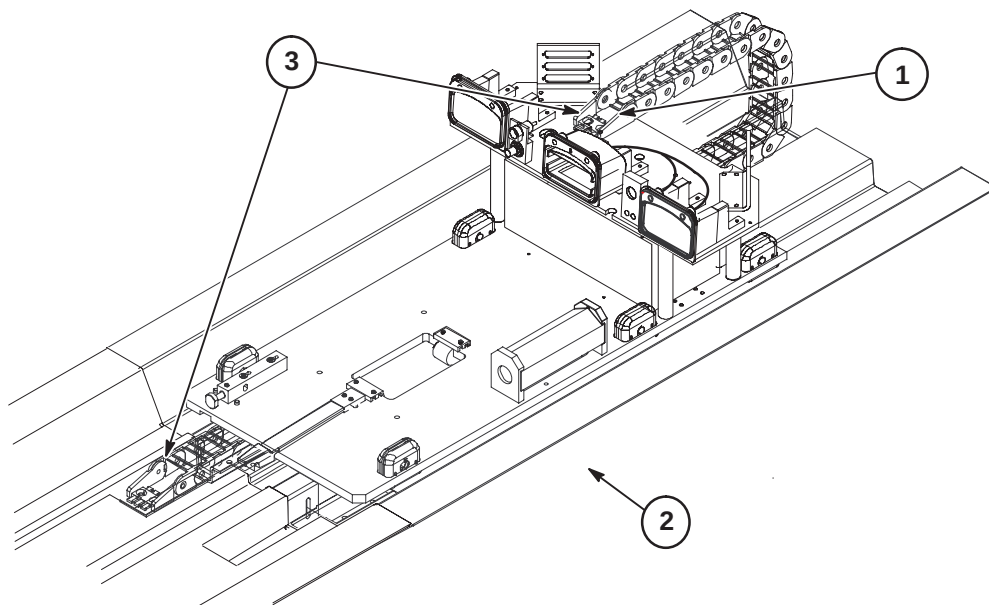
2-5-2 Install New 16-Channel Switch and Image Reject Filter

1. Disconnect cables attached to existing Depopulated 16-Channel Switch
2. Unfasten mounting screws and remove Depopulated 16-Channel Switch.
3. Install new 16-Channel Switch (5109645 for 1.5T or 5109645-2 for 3.0T) and secure in place with previously removed screws.
4. Reconnect cables to new 16-Channel Switch.
5. Attach IRF Mounting Plate (5264778) to Image Reject Filter module (5137348 for 1.5T or 5137348-2 for 3.0T) using four M3 x 12 mm screws (46-312342P24).
6. Attach IRF Module and Mounting Plate assembly to Rear Pedestal using two M6 x 16mm Hex Head Screws (2109866-11)

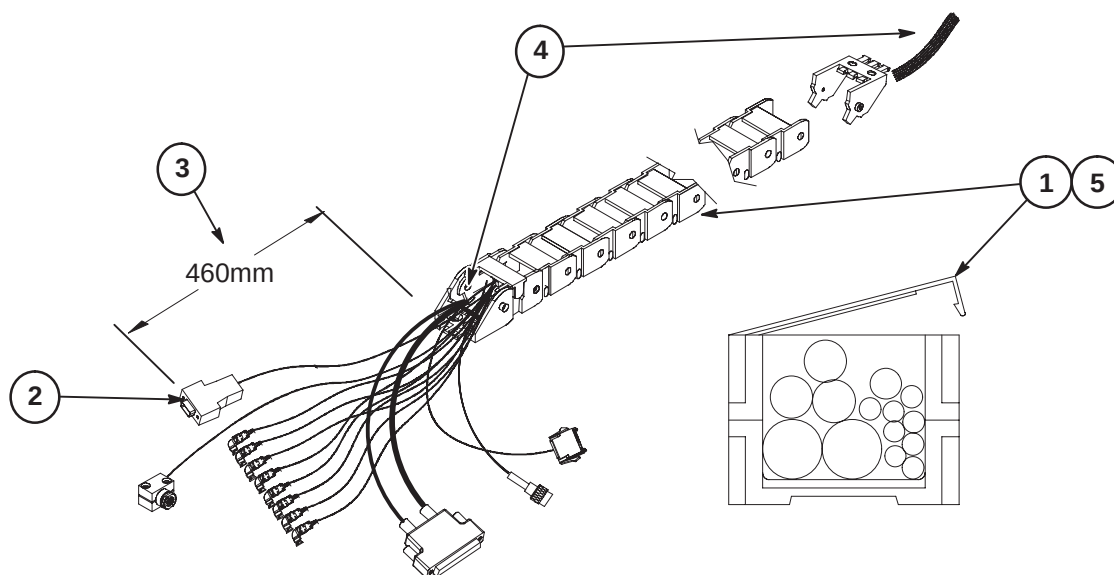


2-5-3 Disconnect Existing Cable Track

1. Disconnect all cables in existing cable track that are connected to LPCA.
2. Disconnect all cables in existing cable track that are connected to modules in Rear Pedestal.
3. Unfasten cable track from LPCA and Bridge. Retain screws for reattaching cable track after upgrade.

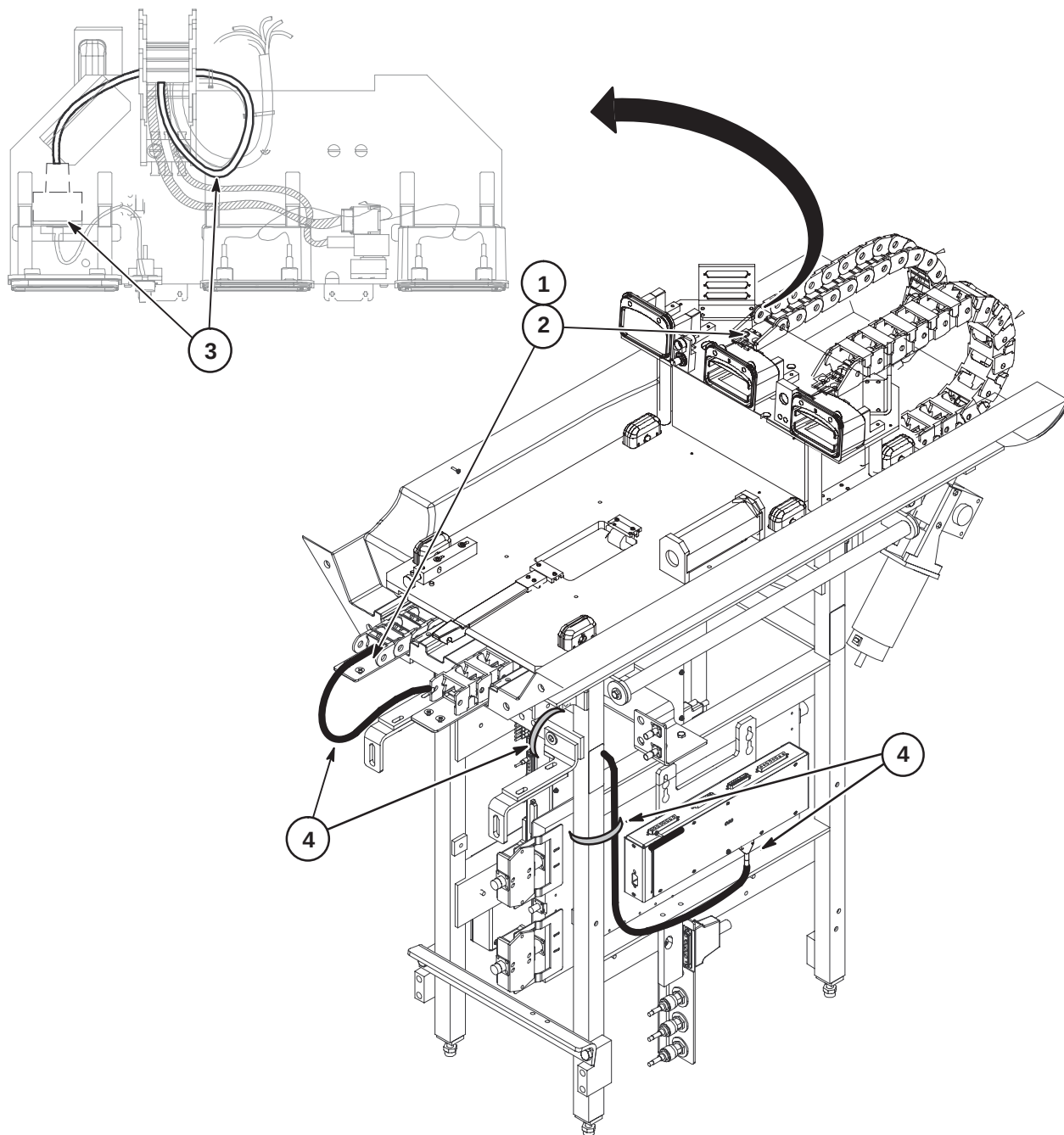
**2-5-4 Install Run 1338 in Existing Cable Track**

1. Lay removed cable track on flat area. Using a screwdriver, open the hinged tops of all the links to access cables in track.
2. Place Run 1338 in cable track.
3. Locate the LPCA connection end of Run 1338 at a distance of 460mm (~18 in.) from the LPCA end of the cable track.
4. Tie-wrap Run 1338 in place.
5. Close the hinged tops of all links.



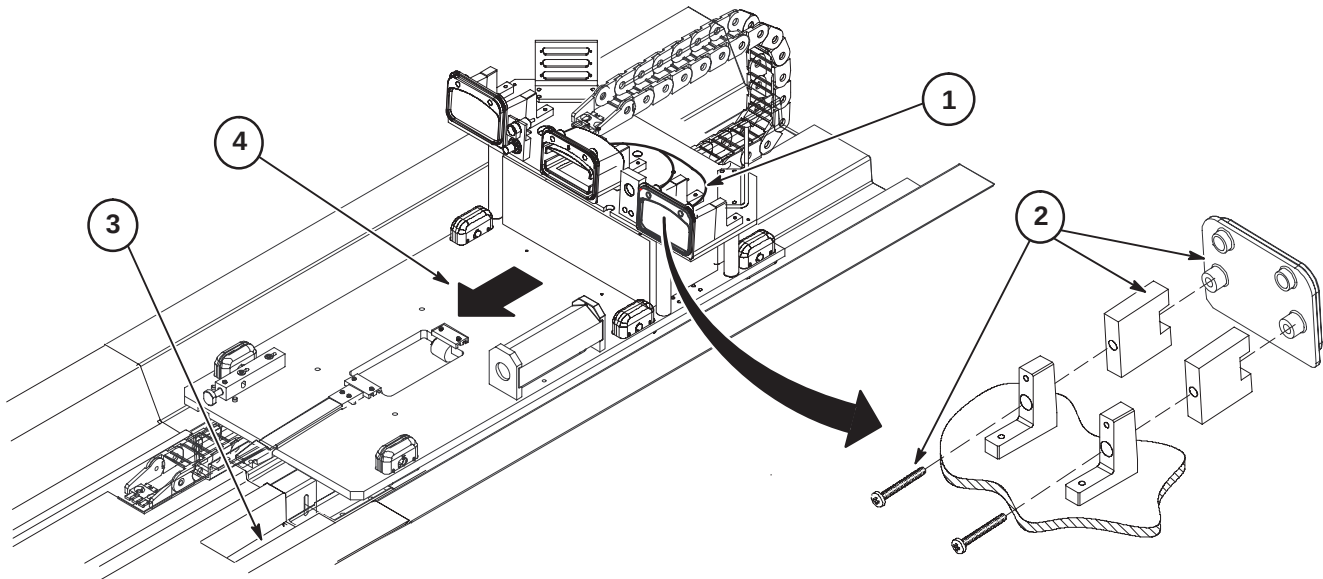
2-5-5 Reconnect Cable Track Cables and Run 1338

1. Fasten cable track to LPCA and Bridge using previously removed screws.
2. Reconnect all cables in existing cable track.
3. Route and connect Run 1338 to J3 in LPCA as shown. J3 is located under shelf.
4. Route and connect Run 1338 to J4 on Image Reject Filter as shown. Secure with tie-wraps as shown.



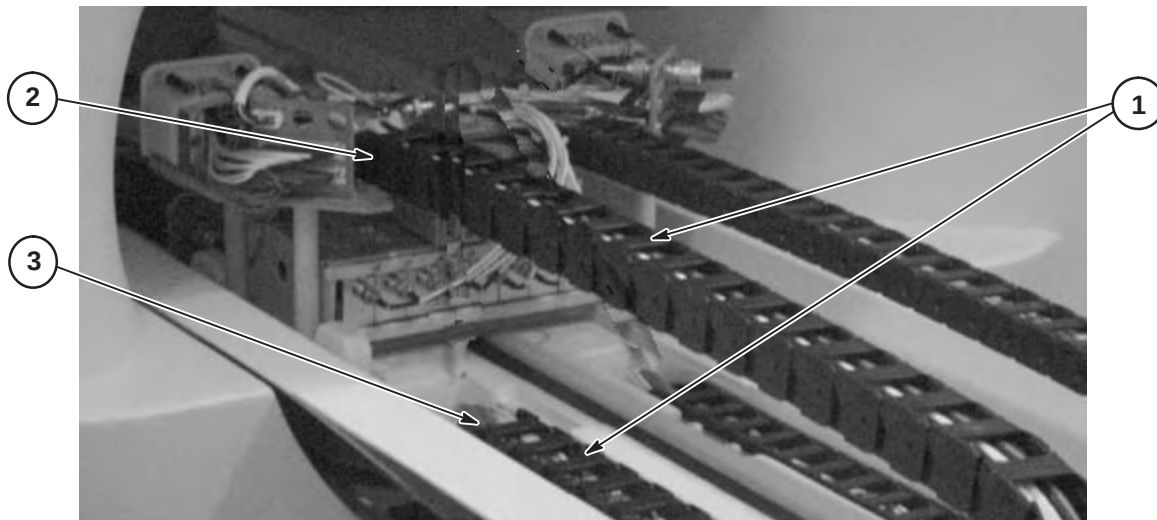
2-5-6 De-Install the B-Port Blanking Plate and the Cable Track Cover Plate

1. Disconnect and retain LED cables from back of Bezel Cover.
2. Unfasten and discard the existing Bezel Cover, Spacer, and M5 x 20mm Screws.
3. Remove and discard Cable Track Opening Cover. Retain screws for installation of new Cable Track.
4. Move LPCA forward into bore so rear of LPCA plate is located just past the cable track opening in the rear of the Front Split Bridge.

**2-5-7 Install Cable Track and Runs 1339, 1340, and 1341**

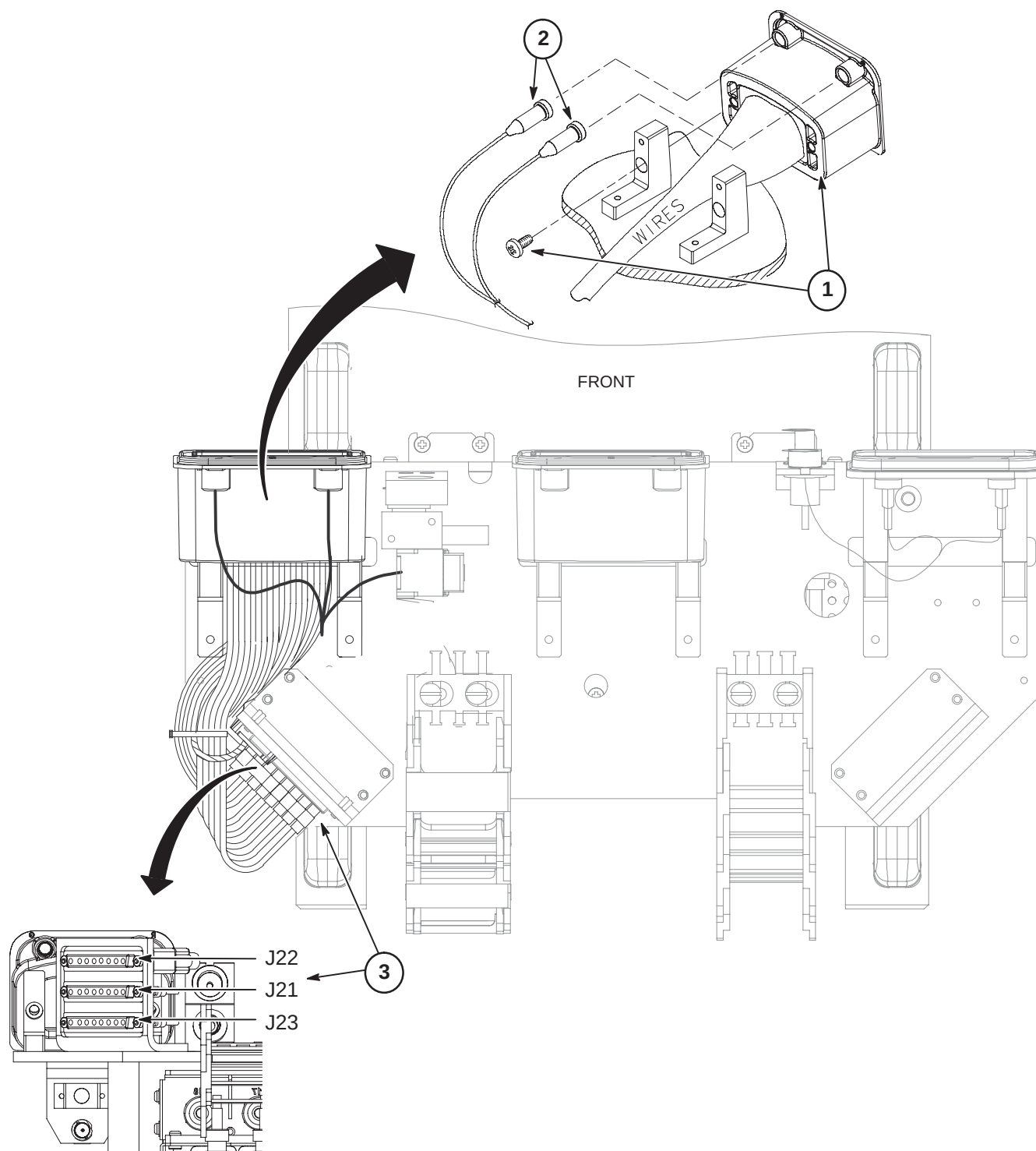
1. Insert the Cable Track assembly into the rear bridge cable track.
2. Remove two screws at location where end of Cable Track attaches to LPCA and apply two drops of LockTite to each screw. Fasten the LPCA end of the cable track to the LPCA with the two screws.
3. Apply two drops of LockTite to the two screws previously removed at the cable track opening in Bridge and attach the other end of the Cable Track.

NOTE: Failure to apply two drops of proper thread locking compound to each cable chain screw before installation could result in screws loosening later and causing spike noise.



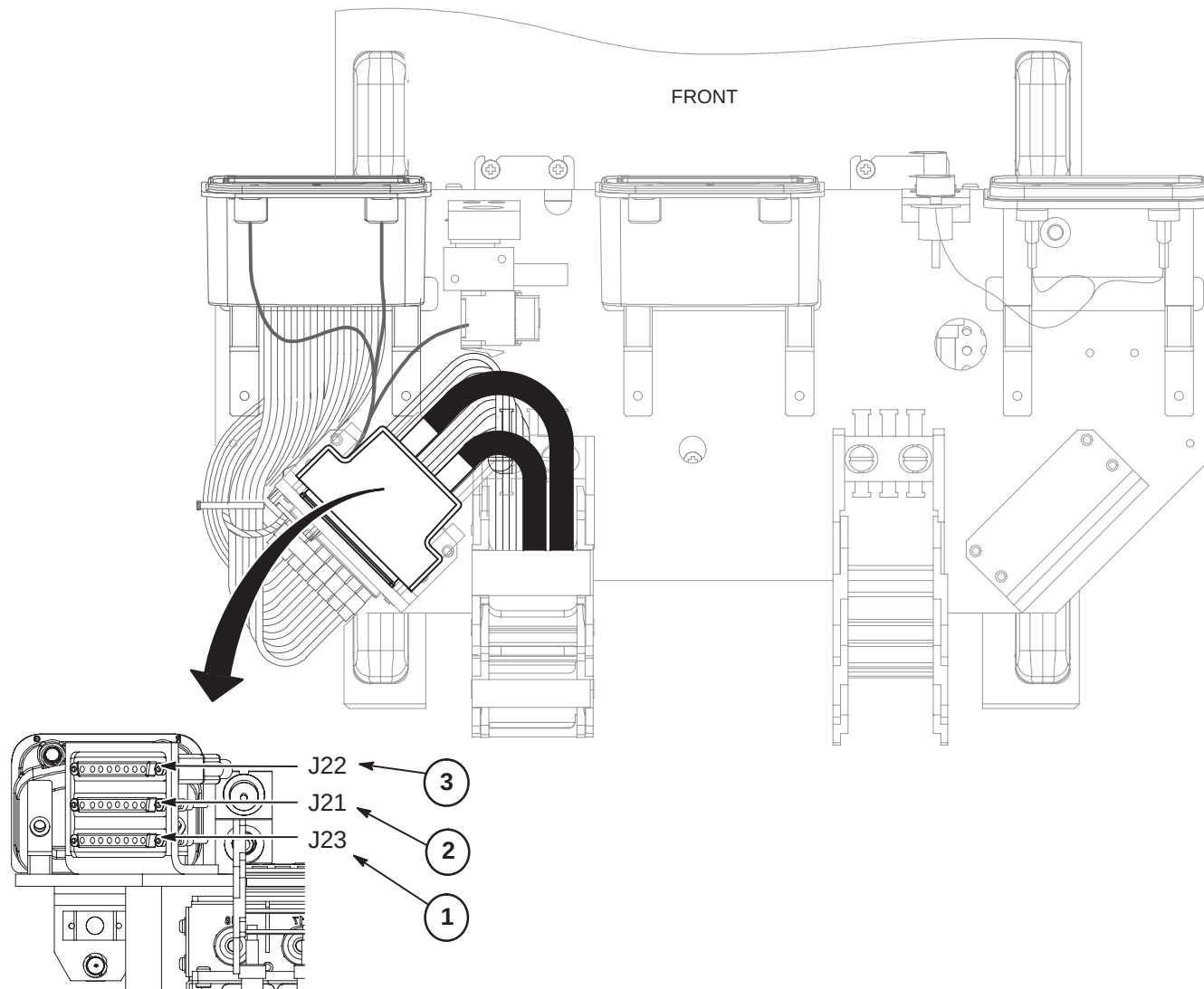
2-5-8 Install B16-Bezel in LPCA

1. Install B16 Bezel and Cable harness (5111020-52) using two M5 x 20mm Brass Screws.
2. Insert the 2 LED's into Bezel.
3. Connect harness to J21, J22, and J23 to bracket as marked using provided nut and standoff on bracket.



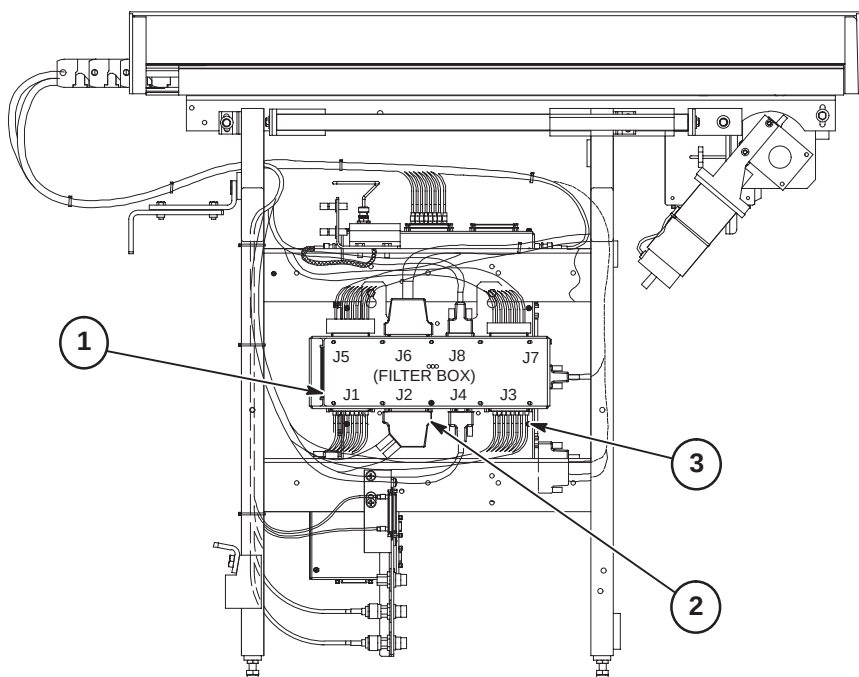
2-5-9 Connect Runs 1339, 1340, and 1341 to B16 Bezel Connectors

1. Connect end of cable run 1339 to J23 on the LPCA bracket.
2. Connect end of cable run 1341 to J21 on the LPCA bracket.
3. Connect end of cable run 1340 to J22 on the LPCA bracket.



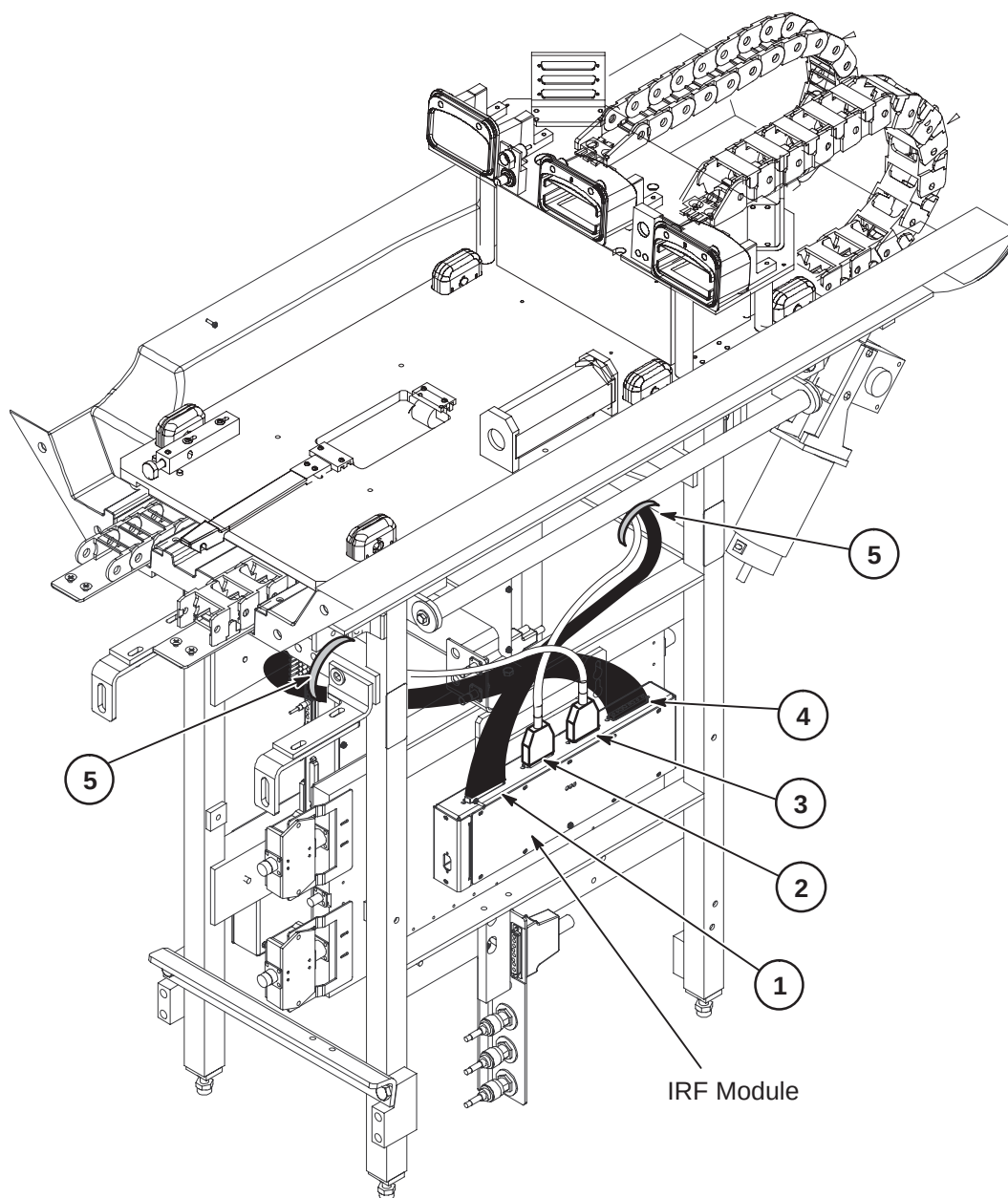
2-5-10 Connect Runs 1339, 1340, and 1341 to IRF Module

1. Connect cable run 1339 to J1 on the Image Reject Filter.
2. Connect cable run 1340 to J2 on the Image Reject Filter.
3. Connect cable run 1341 to J3 on the Image Reject Filter.



2-5-11 Install Runs 1334, 1335, 1336 and 1337

1. Connect one end of Run 1334 as marked to J5 on IRF module. Route along upper back bracket of Rear Pedestal and connect to 16 Channel Switch J22.
2. Connect one end of Run 1335 as marked to J6 on IRF module. Route along upper back bracket of Rear Pedestal and connect to 16 Channel Switch J17.
3. Connect one end of Run 1337 as marked to J8 on IRF module. Route along upper front bracket of Rear Pedestal and connect to 16 Channel Switch J19.
4. Connect one end of Run 1336 as marked to J7 on IRF module. Route along upper front bracket of Rear Pedestal and connect to 16 Channel Switch J24.
5. Dress and tie-wrap Runs to upper front and back brackets of Rear Pedestal.



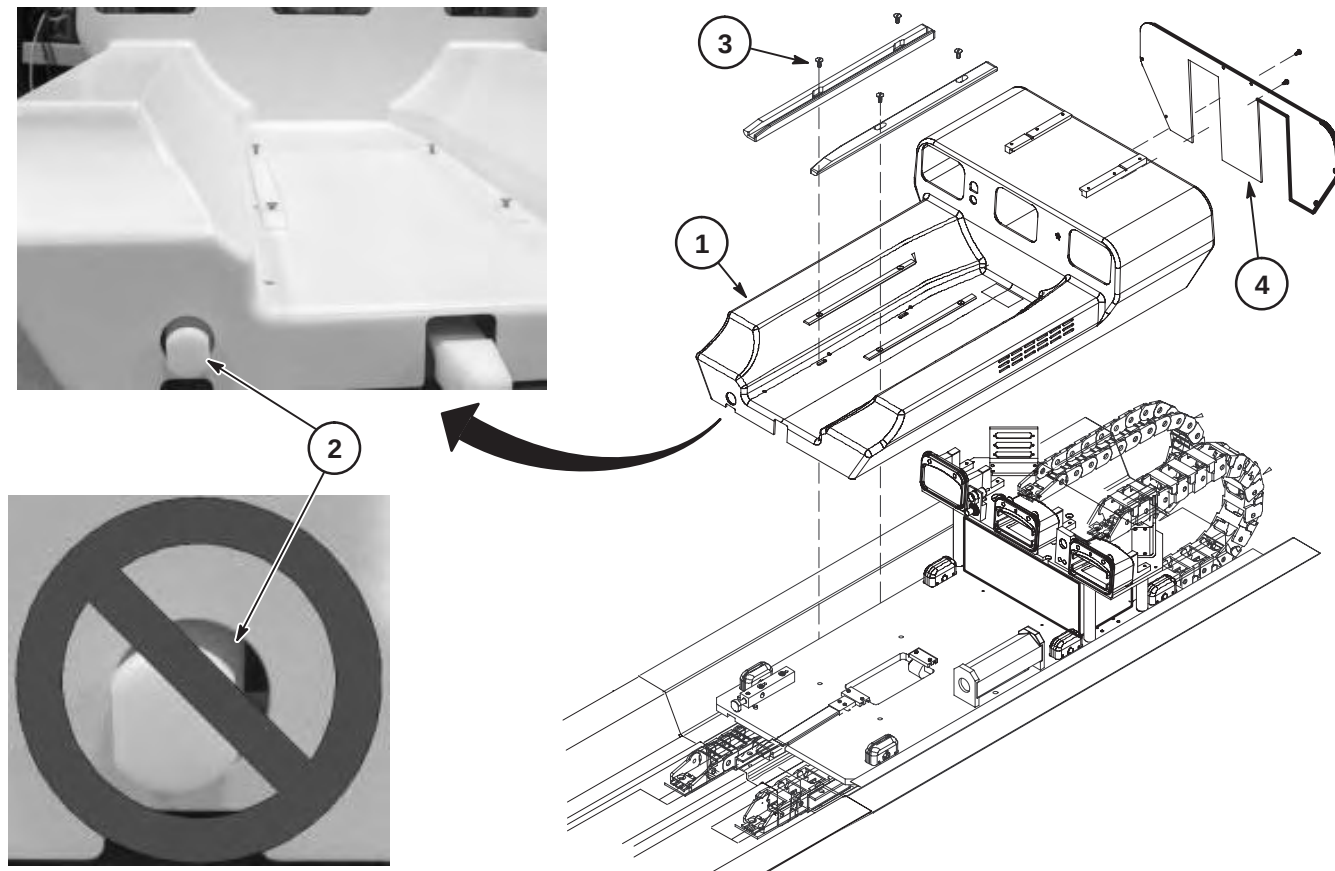
2–5–12 Install Runs 1186, 1201, 1217, and 1329

Refer to applicable cable map in Section 2–2.

1. Connect one end of Run 1186 as marked to J201 on Penetration Panel. Route and connect other end to RFS Cabinet J201.
2. Connect one end of Run 1201 as marked to J201 on Penetration Panel. Route and connect other end to 16–Channel Switch J2.
3. Connect one end of Run 1217 as marked to J78 on Penetration Panel. Route and connect other end to 16–Channel Switch J10.
4. Connect one end of Run 1329 as marked to J78 on Penetration Panel. Route and connect other end to RFS Cabinet J10.

2-6 INSTALL LPCA COVER

1. Carefully place cover on base plate assembly.
2. Center the front knob within the hole on the x-axis. The y-axis does not need to be centered.
3. Once centered, secure the cover to the base plate using four screws previously removed.
4. Attach Rear Cover using 6 screws previously removed.



2-7 INSTALL SOFTWARE

GE Field Engineer to install the 16-Channel option key via eLicense located on the MR Service Methods CD at: Installation//Options/Software Option Install with eLicense Generated Keys

2-8 FINAL SYSTEM CHECKOUT

Run MCR Tool to confirm that all 16 channels are functioning.